# Introduction to Stochastic Calculus

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# Outline


1. Ordinary Calculus
2. Geometric Brownian Motion
3. References

In stochastic calculus, we want to consider integral like

$$\int_0^t f(s)dB(s) \quad \text{or} \quad \int_0^t X(s)dB(s).$$

Since Brownian sample paths are not differentiable, the ordinary calculus intuition will not work because there is no process $B'(t) = \frac{dB(t)}{dt}$, and hence we cannot express

$$\int_0^t g(s)dB(s)$$

as

$$\int_0^t g(s)B'(s)ds.$$

Let $G(t)$ be a differentiable function such that its derivative

$$\frac{dG(t)}{dt} = \lim_{h \to 0} \frac{G(t+h) - G(t)}{h}$$

exists. Then the following three statements are equivalent:

1. $\frac{dG(t)}{dt} = g(t)$ (derivative)
2. $dG(t) = g(t)dt$ ( which is a differential equation)
3. $G(t) = G(0) + \int_0^t g(s)ds$ (which is an integral equation)

Thus one can easily interpret that

$$\int_0^t f(s)dG(s) = \int_0^t f(s)g(s)ds.$$

Suppose

$$f(t) = c\mathbb{I}_{(a,b]}(t) = \begin{cases} c & \text{if } t \in (a,b] \\ 0 & \text{otherwise .} \end{cases}$$

The Riemann integral of this function is

$$\int_0^t f(s)ds = \int_0^t c\mathbb{I}_{(a,b]}(s)ds = \begin{cases} 0 & \text{if } \ t \leq a \\ c(t-a) & \text{if } \ a < t \leq b \\ c(b-a) & \text{if } \ t > b. \end{cases}$$

Let $f$ and $g$ both be functions of $t \in [0, T]$, and consider a partition $0 = t_0 < t_1 < \cdots < t_n = T$, and a corresponding set of intermediate time points values $t_i^*$ such that $t_{i-1} \leq t_i^* \leq t_i$. We assume that $\max |t_i - t_{i-1}| \to 0$ as $n \to \infty$. Then the Riemann-Stieltjes integral of $f$ with respect to $g$ is

$$\int_0^T f(t)dg(t) = \lim_{n\to\infty} \sum_{i=1}^{n} f(t_i^*)[g(t_i) - g(t_{i-1})],$$

if the limit exists.

### Properties

- When $g(t) = t$, we call this a Riemann integral.
- The Riemann-Stieltjes integral exists if either $f$ or $g$ (or both) are differentiable.
- The Riemann-Stieltjes integral exists if either $f$ or $g$ (or both) are of bounded variation.

# The Riemann-Stieltjes integral for simple function

Suppose

$$f(t) = c\mathbb{I}_{(a,b]}(t) = \begin{cases} c & \text{if } t \in (a,b] \\ 0 & \text{otherwise .} \end{cases}$$

The Riemann-Stieltjes integral of this function is

$$\int_0^t f(s)dg(s) = \int_0^t c\mathbb{I}_{(a,b]}(s)dg(s) = \begin{cases} 0 & \text{if } \ t \leq a \\ c(g(t) - g(a)) & \text{if } \ a < t \leq b \\ c(g(b) - g(a)) & \text{if } \ t > b. \end{cases}$$

**Definition (Basic Stochastic Process)**

The process $X$ is called a basic stochastic process if $X$ admits the following representation

$$X_t(\omega) = C(\omega)\mathbb{I}(a,b](t), \;\; a < b \in \mathbb{R}$$

and $C$ is a $\mathcal{F}_a$ measurable random variable.

The stochastic integral of $X$ with respect to the Brownian motion is defined as

$$\left(\int_0^t X(s)dB(s)\right)(\omega) = \begin{cases} 0 & \text{if} \;\; 0 \leq t \leq a \\ C(\omega)(B(t,\omega) - B(a,\omega)) & \text{if} \;\; a < t \leq b \\ C(\omega)(B(b,\omega) - B(a,\omega)) & \text{if} \;\; t > b. \end{cases}$$

# Chain Rule in Ordinary Calculus

For $f : \mathbb{R} \to \mathbb{R}$ and $x : [0, \infty) \to \mathbb{R}$, which are continuously differentiable, we know that

$$\frac{df(x(t))}{dt} = f'(x(t))\frac{dx(t)}{dt}$$

or in differential form

$$df(x(t)) = f'(x(t))dx(t).$$

We prove this by using Taylor series expansion such that

$$df(x(t)) = f'(x(t))dx(t) + \frac{1}{2!}f''(x(t))dx(t)dx(t) + \cdots .$$

Here df(x(t)) denote the infinitesimal increment of $f(x(t))$. Note that dx(t)dx(t) corresponds to the quadratic variation of the function $x(\cdot)$.

Consider the Ito process $X(\cdot)$ given by

$$dX(t) = Q(t)dt + P(t)dB(t).$$

For $f : \mathbb{R} \to \mathbb{R}$ be a function which has derivatives of all orders. Then using Taylor series expansion of $f$, we get

$$df(X(t)) = f'(X(t))dX(t) + \frac{1}{2!}f''(X(t))dX(t)dX(t) + \frac{1}{3!}f'''(X(t))dX(t)dX(t)dX(t) + \cdots,$$

since variation of $X(\cdot)$ of order 3 and more are zero, we get

$$df(X(t)) = f'(X(t))dX(t) + \frac{1}{2!}f''(X(t))dX(t)dX(t).$$

Note that, we have $dX(t)dX(t) = P(t)^2dt$, which leads to

$$df(X(t)) = [Q(t)f'(X(t)) + \frac{1}{2}P(t)^2f''(X(t))]dt + P(t)f'(X(t))dB(t).$$

**Proposition**

*For an Ito process $dX(t) = P(t)dB(t) + Q(t)dt$, the Ito's formula is given by*

$$\begin{aligned} df(t, X(t)) &= f_t(t, X(t))dt + f_x(t, X(t))P(t)dB(t) \\ &\quad + f_x(t, X(t))Q(t)d(t) + \frac{1}{2} f_{xx}(t, X(t))P^2(t)dt. \end{aligned}$$

**Proof.**

Using Taylor series expansion, we have

$$df(t, X(t)) = f_t(t, X(t))dt + f_x(t, X(t))dX(t) + \frac{1}{2} f_{xx}(t, X(t))dX(t)dX(t).$$

We have $dX(t)dX(t) = P(t)^2 dt$, which leads to

$$\begin{aligned} df(t, X(t)) &= f_t(t, X(t))dt + f_x(t, X(t))P(t)dB(t) \\ &\quad + f_x(t, X(t))Q(t)d(t) + \frac{1}{2} f_{xx}(t, X(t))P^2(t)dt. \end{aligned}$$

□

# Stock price under the Bachelier model

- Under Bachelier model stock price is governed by the equation $dS(t) = \sigma dB(t)$.
- It follows, $S_t = S_0 + \sigma B_t$, where $\sigma$ is volatility term.
- Thus, $S_t - S_{t-1} = \sigma(B_t - B_{t-1})$
- Hence $S_t - S_{t-1} \sim N(0, \sigma^2)$.
- Further, $S_t - S_{t-1}, t = 1, 2, \ldots$ are independent.
- $\hat{\sigma} = sd(S_t - S_{t-1})$.

# Geometric Brownian motion

- The geometric Brownian motion is a solution of following SDE

$$dS(t) = \mu S(t)dt + \sigma S(t)dB(t).$$

- Choose $f(x,t) = \ln x$, which implies by Ito's formula

$$d\ln(S(t)) = 0 + \frac{1}{S(t)}P(t)dB(t) + \frac{1}{S(t)}Q(t)dt + \frac{1}{2}\frac{-1}{S^2(t)}P^2(t)dt.$$

- By substituting $P(t) = \sigma S(t)$ and $Q(t) = \mu S(t)$, it follows

$$\begin{aligned} d\ln(S(t)) &= \frac{1}{S(t)}\sigma S(t)dB(t) + \frac{1}{S(t)}\mu S(t)dt + \frac{1}{2}\frac{-1}{S^2(t)}\sigma S^2(t)dt \\ &= \sigma dB(t) + (\mu - \frac{\sigma^2}{2})dt. \end{aligned}$$

- Which in turn implies $S(t) = S(0)e^{(\mu - \frac{\sigma^2}{2})t + \sigma B(t)}$.

# Estimation of parameters of GBM

- We have $S(t) = S(0)e^{(\mu - \frac{\sigma^2}{2})t + \sigma B(t)}$.
- Thus the log-returns $r(t) = \log\left(\frac{S(t)}{S(t-1)}\right) = (\mu - \frac{\sigma^2}{2}) + \sigma(B(t) - B(t-1))$.
- Hence $r(t)$ is normal with mean $\left(\mu - \frac{\sigma^2}{2}\right)$ and variance $\sigma^2$.
- Further $r(t), t = 1, 2, \ldots, T$. are independent, since these are modeled by the increments of Brownian motion with drift.
- Thus, the estimate of $\sigma^2$ is sample variance i.e. $\widehat{\sigma^2} = \frac{1}{T-1}\sum_{t=1}^{T}(r(t) - \overline{r(t)})^2$.
- The estimate of population mean by taking sample mean implies $\widehat{\left(\mu - \frac{\sigma^2}{2}\right)} = \overline{r(t)}$. Hence $\hat{\mu} = \overline{r(t)} + \frac{\widehat{\sigma^2}}{2}$.
- Hence under the GBM model, the parameters can be estimated by daily log returns of the equity.

The GBM is given by

$$S(t) = S(0)e^{(\mu - \frac{\sigma^2}{2})t + \sigma B(t)}.$$

### Steps in simulation of GBM

- Discretize the interval $[0, T]$ into uniform intervals of length $\Delta t$ each.
- Generate $N = T/\Delta t$ normal random numbers with mean 0 and variance $\sigma^2 \Delta t$.
- Take the cumulative sum of these random numbers. Call this vector as $p_1$.
- Calculate $\left(\mu - \frac{\sigma^2}{2}\right) j\Delta t, j = 1, 2, \ldots, N$. Call this vector $p_2$.
- Now GBM vector is defined by $S(t) = e^{p_1 + p_2}$.

Compute the stochastic differentials $dY$ when

a. $Y(t) = \exp(\beta t)$
b. $Y(t) = \exp(\beta B(t))$
c. $Y(t) = \exp(\beta X(t))$ and $dX(t) = \mu dt + \sigma dB(t)$.
d. $Y(t) = 1/X(t)$ where $dX(t) = \alpha X(t)dt + \sigma X(t)dB(t)$.

Show that the process defined by $Y(t) := t^2 B(t)^3,\ t \geq 0$ satisfies

$$dY(t) = (2Y(t)/t + 3(t^4 Y(t))^{1/3})dt + 3(tY(t))^{2/3} dB(t),\ \ Y(0) = 0.$$

**Solution:** Consider the function $f(t, x) = t^2 x^3$ and apply the Ito formula.

# Useful Properties of the Ito Integral

- $\mathbb{E}\int_0^t X(s)dB(s) = 0.$
- Ito isometry property

$$\mathbb{E}\left[\int_0^t X(s)dB(s)\right]^2 = \int_0^t \mathbb{E}[X(s)^2]ds.$$

- $$\mathbb{E}\left[\int_0^t X(s)dB(s)\int_0^t Y(s)dB(s)\right] = \int_0^t \mathbb{E}[X(s)Y(s)]ds.$$

# Solving Ornstein-Uhlenbeck SDE

- The OU process satisfies the stochastic differential equation

$$dX(t) = -\lambda X(t)dt + \sigma dB(t),$$

- For an Ito process $dX(t) = P(t)dB(t) + Q(t)dt$, the Ito's formula is given by

$$\begin{aligned} df(t, X(t)) &= f_t(t, X(t))dt + f_x(t, X(t))P(t)dB(t) \\ &\quad + f_x(t, X(t))Q(t)d(t) + \frac{1}{2} f_{xx}(t, X(t))P^2(t)dt. \end{aligned}$$

- Consider $X(t) = e^{-\lambda t}Y(t)$ which implies $Y(t) = e^{\lambda t}X(t)$. Using Ito formula for $f(x, t) = e^{\lambda t}x$, we have

$$dY(t) = \lambda e^{\lambda t}X(t)dt + e^{\lambda t}\sigma dB(t) - \lambda e^{\lambda t}X(t)dt = e^{\lambda t}\sigma dB(t).$$

  Thus

$$Y(t) = Y(0) + \sigma \int_0^t e^{\lambda s}dB(s) \implies X(t) = X(0)e^{-\lambda t} + \sigma e^{-\lambda t} \int_0^t e^{\lambda s}dB(s).$$

# Mean and Variance of OU Process

- We have $X(t) = X(0)e^{-\lambda t} + \sigma e^{-\lambda t} \int_0^t e^{\lambda s} dB(s)$.
- $\mathbb{E}[X(t)] = X(0)e^{-\lambda t}$.
- The variance is given by

$$
\begin{aligned}
Var(X(t)) = \mathbb{E}(X(t) - \mathbb{E}X(t))^2 &= \mathbb{E}\left[\sigma e^{-\lambda t} \int_0^t e^{\lambda s} dB(s)\right]^2 \\
&= \sigma^2 e^{-2\lambda t} \int_0^t e^{2\lambda s} ds \\
&= \sigma^2 e^{-2\lambda t} \left[\frac{e^{2\lambda t}}{2\lambda} - \frac{1}{2\lambda}\right] \\
&= \frac{\sigma^2}{2\lambda}(1 - e^{-2\lambda t}).
\end{aligned}
$$

# Some References


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**THANK YOU**